MO3TAMAD
Certified assessments

Department of Computer Science

Department of Computer Information Systems

KING ABDULLAH II SCHOOL OF INFORMATION TECHNOLOGY

THE UNIVERSITY OF JORDAN

JANUARY 2023

Abdallah Almaharmeh	**0193598**
Haroun Salem	0195706
Saif allah Hamdan	0198502

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ABSTRACT

It has become difficult to determine the level of individuals in a profession or skill, you can obtain certificates without making the required effort, and methods of cheating have become many, especially in online exams.

MO3TAMAD is an idea that arose to measure the level of technical skills of individuals by holding online adaptive exams that are designed to ensure the integrity and transparency of the exams. The examinee chooses a technical skill; a question bank is already prepared for it; and at the start of the exam, the site monitors the sound, records the video, and tracks opened applications on the device so that it tries as much as possible to reduce the student's ability to cheat, and through the idea of adaptive exams, MO3TAMAD can determine the student's level with the least number of questions.

In this way, we can give accredited certificates to the examinees and determine their level accurately so that they can use these certificates to prove their capabilities when applying for available opportunities such as work or internships or share them with friends on LinkedIn or other platforms.

MO3TAMAD aims to reduce the time and effort required to prove your abilities in skill, so that you can at any time or place prove your level in technical skill and, at the same time, give the real level to each examinee through an official certificate from MO3TAMAD that he will not take if he tries to cheat, and so we try even a little harder to give opportunities to those who deserve them by proving the capabilities of individuals with professionalism, integrity, and transparency.

ACKNOWLEDGMENTS

After thanking God Almighty for all His blessings, we would like to extend our thanks and gratitude to the professors supervising us in this project, for their continuous support and constant guidance, Dr. Bilal Abu Saleh, Dr. Reem Al-Fayez, and Dr. Saleh Al- sharaeh.

This is gratitude to all those who supported us to obtain a bachelor's degree, to our families, friends, and loved ones, to our teachers and professors, to King Abdullah II College for Information Technology and all its distinguished professors and teachers, to the University of Jordan, this scientific edifice to which we are proud to belong, and to this dear country that we hope to see at the forefront of other countries.

Thank you very much, all.

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1.0 CHAPTER ONE: INTRODUCTION

1.1 Preamble

It has become difficult to determine the level of individuals in a profession or skill, you can obtain certificates without making the required effort, and methods of cheating have become many, especially in online exams.

MO3TAMAD is an idea that arose to measure the level of technical skills of individuals by holding online adaptive exams that are designed to ensure the integrity and transparency of the exams. The examinee chooses a technical skill; a question bank is already prepared for it; and at the start of the exam, the site monitors the sound, records the video, and tracks opened applications on the device so that it tries as much as possible to reduce the student's ability to cheat, and through the idea of adaptive exams, MO3TAMAD can determine the student's level with the least number of questions.

In this way, we can give accredited certificates to the examinees and determine their level accurately so that they can use these certificates to prove their capabilities when applying for available opportunities such as work or internships or share them with friends on LinkedIn or other platforms.

MO3TAMAD aims to reduce the time and effort required to prove your abilities in skill, so that you can at any time or place prove your level in technical skill and, at the same time, give the real level to each examinee through an official certificate from MO3TAMAD that he will not take if he tries to cheat, and so we try even a little harder to give opportunities to those who deserve them by proving the capabilities of individuals with professionalism, integrity, and transparency.

1.2 Project Motivation

Giving opportunities to those who deserve them, maintaining integrity and transparency, and not wasting resources, all these goals and others are the seeds from which MO3TAMAD grew. MO3TAMAD's goal is to provide an effective tool that allows employers to predict the capabilities of those who apply for job openings, ensuring that opportunities go to those who deserve them. At MO3TAMAD, we aim to make our certificates the focus of companies' accreditation, so that we offer real exams that measure the students' level and are free of fraud and deception.

In addition to the above, ease and facilitation are among the most important motives for establishing MO3TAMAD so that individuals can judge their level of a specific technical skill from home at any time convenient for them, without the need to go to an authorized center and pay amounts that may not suit them.

Thus, MO3TAMAD aims to facilitate the process of conducting exams while ensuring their credibility in a modern and pioneering examination system that keeps pace with this era.

1.3 Problem Statement

Testimonials do not measure real experiences; they may be fake, and the individual may obtain them through fraud and deception. Traditional examination methods are expensive and complex. These problems were a reason to start MO3TAMAD to form a revolution in the world of virtual assessments in line with the twenty-first century.

1.4 Project Aim and Objectives

MO3TAMAD aims to:

- giving individuals accredited, transparent, and fair IT certificates to assess their levels.
- Facilitating the recruitment process by providing reliable certificates that companies can adopt instead of wasteful interviews.
- making the process of obtaining certificates easier than the traditional approved and costly method.
- Reduce the number of workers by using artificial intelligence algorithms to detect fraud and cheating.
- Introducing smart exams designed to adjust the level of difficulty based on the answers provided instead of long, cumbersome exams

1.5 Project Scope

Boundaries:

- Examinees who want to determine their level in a skill and take an accredited certificate.
- Companies and employers who want a reliable way to determine the level of job applicants.
- University students and professors.

Deliverables:

- Adaptive IT assessments with Less time and more accuracy.
- A reliable system that detects fraud.
- Certified certificates have the least effort, the fastest time, and very great credibility and trust.

1.6 Project Software and Hardware Requirements

1.6.1 Software Requirements

Software	Requirement
Operating System	Windows 7, 8.1 or higher operating systems, MacOS 10.12, Ubuntu LTS releases 18.04 or late, iOS 7.1.2 or higher, Android devices with Android 4.2+.
Browser	Internet Explorer 8.0 (64-bit Internet Explorer 7.0 is not supported), Firefox 3.0, Safari 4.1.2, and Google Chrome 11.0, all these browsers in the minimum versions.
Development tools	Visual studio code, Golang, Figma (diagram), Docker, TablePlus, Postman.
Database Software	Apache Server 1.7.1 or Higher included MySQL

Table 1 - Software Requirements

1.6.2 Hardware Requirements

Hardware	Requirement
Computer	Any computer with a web browser
Memory (RAM)	At least 2GB, preferably higher.
Hard Disk	At least 10 GB
CPU	Core i3 - 1480 MHz Pentium minimum, V - 1 GHz or higher recommended

Table 2 - Hardware Requirements

1.7 Project Limitations

- Time: The initial website should be ready by 15 June 2023 for users to use.
- Internet: One limitation of our project is that we need an internet connection to access the website.
- Supported languages: Our website is in English; we haven't worked on an Arabic or the other languages yet.
- We fear that the first versions of the site will not be able to detect all forms of fraud due to the lack of reliable datasets, so we will make future plans and upgrade our system.

1.8 Project Contributions (Expected Output)

This project will provide a website to issue accredited certificates while ensuring that all the objectives mentioned in 1.4 are achieved.

The output of the project will be a fully functional website. It will contain a front end with which the users of the website can interact. It will also contain a back-end dashboard for admin usage.

1.9 Project Schedule

Task	Description	Work began	Work end	Duration (days)	Dependency
Task (1)	Project Name, Brainstorming	20 October 2022	25 October 2022	5 days	-
Task (2)	Documentation	25 October 2022	15 January 2023	80 days	-
Task (2.1)	Introduction	25 October 2022	1 November 2022	5 days	-
Task (2.2)	Related existing systems	2 November 2022	4 November 2022	2 days	-
Task (2.3)	System requirements engineering and analysis	5 November 2022	5 December 2022	30 days	-
Task (2.4)	System design	6 December 2022	15 January 2023	40 days	Task (2.3)
Task (3)	System implementation	1 February 2023	1 April 2023	60 days	Task (2)
Task (4)	System testing and installation	1 April 2023	15 April 2023	15 days	Task (3)
Task (**5**)	Conclusion and future work	15 April 2023	1 May 2023	15 days	Task (4)
Task (**6**)	Submission	4 June 2023			Task (5)

Table 3 - Project Schedule

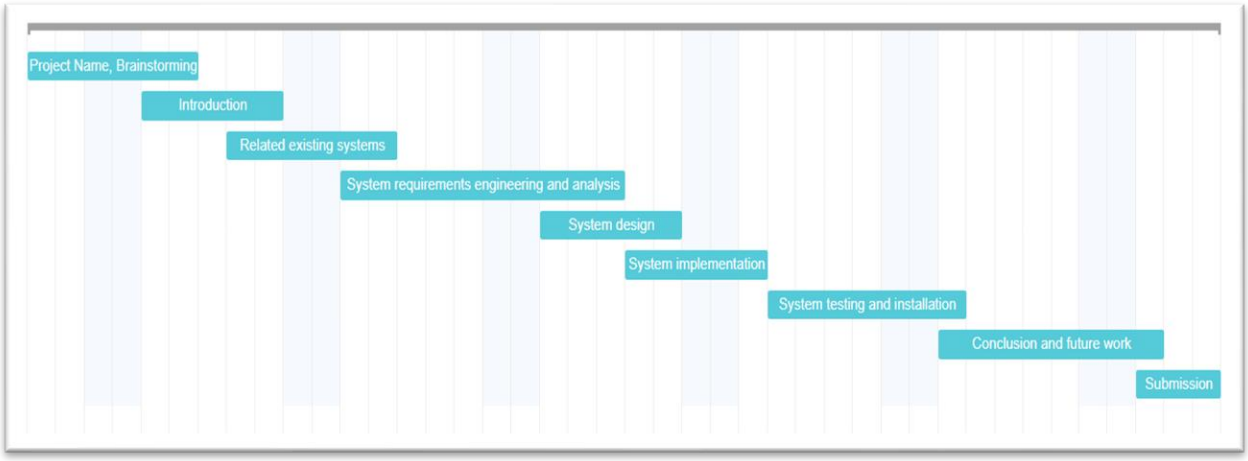

Figure 1 - Gantt Chart

1.10 Report Outline

The project is organized as follows:

- Chapter one: Shows a preamble in section 1.1, the project motivation is elaborated in Section 1.2, in addition, the problem statement is stated in Section 1.3, Section 1.4 lists the project aim and objectives, whereas the project scope is identified in section 1.5, Section 1.6 provide the project software and hardware requirement and Section 1.7 highlights the project limitation, furthermore, project expected output is addressed in Section 1.8, in section 1.9 the project is scheduled, and the report outline is finally present in Section 1.10.

- Chapter Two: Shows an introduction in Section 2.1, the existing system is elaborated in Section 2.2, in addition, the overall problems of existing system are stated in Section 2.3, Section 2.4, overall solution approach, whereas the summary is identified in Section 2.5

- Chapter Three: shows an introduction in Section 3.1, feasibility study is elaborated in Section 3.2, in addition, and Requirements Elicitation Techniques in Section 3.3, the targeted users are stated in Section 3.4, Section 3.5 shows Functional requirements definition and functional requirement specification is identified in Section 3.6, Section 3.7 highlights the nonfunctional requirements, scenarios are stated in section 4.8, furthermore, the summary output is addressed in Section 3.9

- Chapter Four: Shows an introduction in Section 4.1, Context Diagram elaborated in Section 4.2, the Data Flow Diagram (DFD) in Section 4.3, Section 4.4 provide the Entity Relationship Diagram, in addition, the UML, Sequence Diagram is stated in Section 4.5, Section 4.6 include UML Sequence Diagram, Section 4.7 provides UML Class Diagram and Graphical User Interface design in 4.8

2.0 CHAPTER TWO: RELATED EXISTING SYSTEMS

2.1 Introduction

In this chapter we will talk about existing systems that share similarities with our system, the overall problems with these systems and the suggested solutions.

2.2 Existing Systems

- **w3schools:** w3schools is one of the best well-known resources to learn programming languages, they offer a certification exam for every language/tool they teach on their platform.
- **Udemy:** it's a platform that offers courses and in the end of the course they reward the learner with a certification for the completion of the course, a lot of platforms apply this approach like AWS, Google, Microsoft, and Coursera.
- **Duolingo:** it's a platform focused on teaching English, they offer a unique service to take an English Test equal to TOFOL and IEITS for a lower price and you can take it anytime anywhere for 50$, over it's been accepted by over 4000 institutions.

2.3 Overall Problems with Existing Systems

As we have noticed there are no existing systems that specifically offer a trusted certificate for the IT industry, most of the platforms provide a certificate for the languages they teach only and some of them offer it at high prices that make it hard for students to buy, we also noticed that companies like Udemy and Coursera do not trust some of the certificates because their certificates don't mean the student is good at what he does, and the learner can get it by copying projects/assignment that has posted in the internet, Duolingo, for example, offer a great trusted certificate but the problem is it's just for English.

2.4 Overall Solution Approach

The overall solution we figured it out is to develop a platform that provides a trusted IT certificates for its candidates with a reasonable price, we will cover all kind of languages, tools and anything related to IT industry, we will make sure that our exams test the candidate very well in his knowledge, secure by developing algorithms and techniques to prevent candidate from cheating.

2.5 Summary

Finally in this chapter we noted the important issues with the current existing systems and discussed the possible solutions for them

3.0 CHAPTER THREE: SYSTEM REQUIREMENTS ENGINEERING AND ANALYSIS

3.1 Introduction

This chapter includes the system requirements and system analysis with a full description of the system including the definition and the requirements specification.

3.2 Feasibility study

3.2.1 Technical Feasibility

The Project is a browser-based application. Resources that are required for this project, include:

- Programming device
- Hosting space
- Programming tools
- Programming individuals

The main technologies and tools that are associated with this project are:

- React
- MySQL
- Golang
- Docker
- Linux

Each of the technologies are freely available and the technical skills required are manageable. From the study above, the project is technically feasible.

3.2.2 Operational Feasibility

The website will provide technologies that prevent fraud while testing users, which guarantees an accredited certificate with an actual level of knowledge, which makes users and companies depend on these certificates in employment and proving skills.

3.2.3 Economic Feasibility

- **Development Costs:**

Personal costs:

Employee type	Total cost
Three Programmers (160h 12JD/h)	5760 JD
System Analyst (20h 10JD/h)	200 JD

Table 4 - Personal costs

Hardware's and Software's cost:

Hardware/Software	Cost
3 Computers	3000 JD
Server Development	1000 JD
DBMS software	200 JD

Table 5 - Hardware's and Software's cost

Total Development costs: 4200 + 5960 = 10160 JD

- **Operating costs:**

- Employee:

  1 programmer (100 h 12/hr.) = 1200 JD

- Expenses:

  1 Server maintenance agreement (400 JD)

  1 Software maintenance agreement (300JD)

  Total 700 JD

  Operating cost total in one year: 700+1200 = 1900JD

- Benefits:
  a) Tangible Benefits: Benefits are quantifiable and measurable; it includes reduced costs and increased revenue.
     - Reduce number of offline exams that needs rent costs and employee form management.
     - Increase number of attendance of exams. (1000 JD)
     - Error reduction (350 JD)
     - Increased speed of activity (400 JD)
     - Flexibility (400 JD)

  Total: 1000 + 350 + 400 + 400 = 2150

b) Intangible benefits: Benefits that cannot be easily measured in money and with certainty that resulted from information forms.

Cash flow	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Cost	(5960)	0	0	0	0	0
Operating Cost	0	(1900)	(3500)	(5833)	(6350)	(7350)
Benefits	0	2150	3000	4113	5557	7000
Discount Rate (10%)	1	0.9091	0.8264	0.7513	0.6830	0.6209
Time Adjusted Cost	(5000)	(2863)	(3594)	(4019)	(4337)	(4563)
Time Adjusted Benefits	0	2150	4875	5183	5395	5526
Cumulative Time Adjusted Cost	(5000)	(7684)	(11278)	(15297)	(19634)	(24197)
Cumulative Time Adjusted Benefits	0	4454	9329	14512	19907	25433
Cumulative Time Adjusted (Cost& Benefits)	(5000)	(3230)	(1949)	(785)	273	1236

Table 6 - Payback analysis

3.3 Requirements Elicitation Techniques

3.3.1 Questionnaire

We came up with a few questions to help us interact with people who might benefit from our project and help us see the amount of need for our project, the following are the questions and their responses:

Are you interested in measuring your technical skills and obtaining an accredited certificate that proves your level?

84 responses

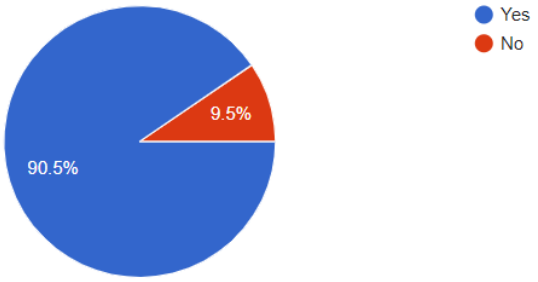

Figure 2 - Questionnaire Question 1

Have you ever studied a course on one of the sites that provide courses "Udemy - Coursera - etc."?

84 responses

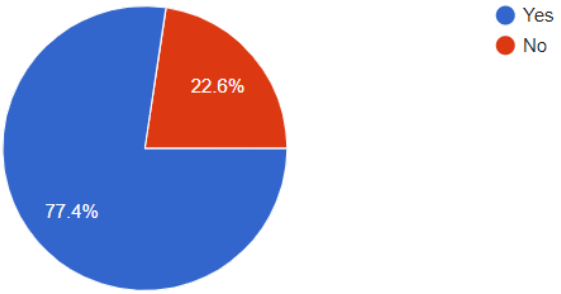

Figure 3 - Questionnaire Question 2

Do you think that the certificates provided as a result of the courses taken online measure the real benefit of the student?

84 responses

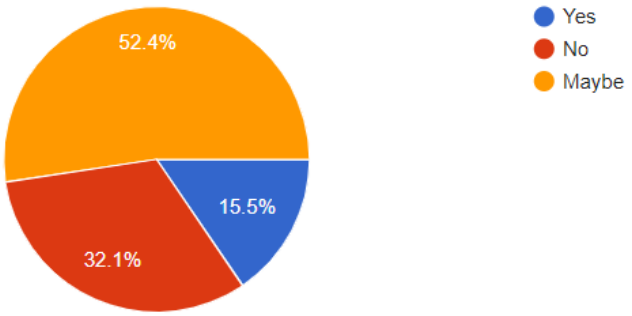

Figure 4 - Questionnaire Question 3

Do you think that electronic exams are fair and free from cheating?

84 responses

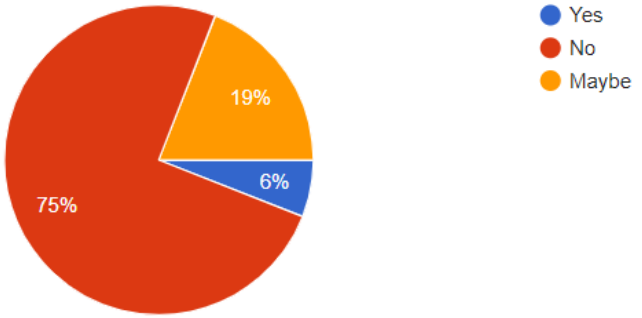

Figure 5 - Questionnaire Question 4

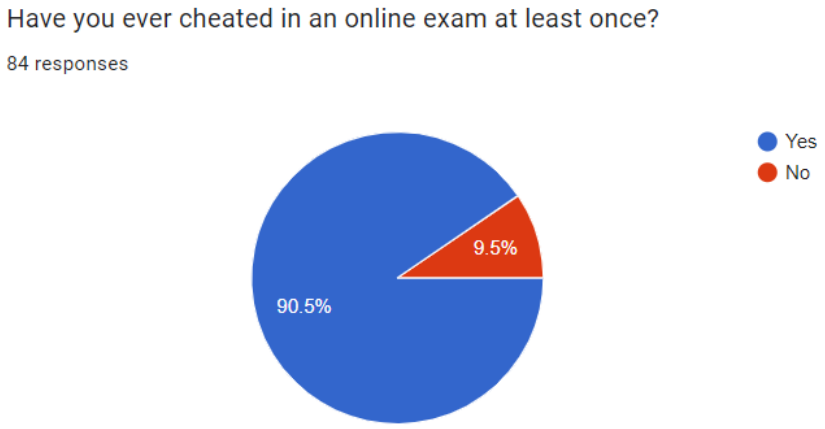

Figure 6 - Questionnaire Question 5

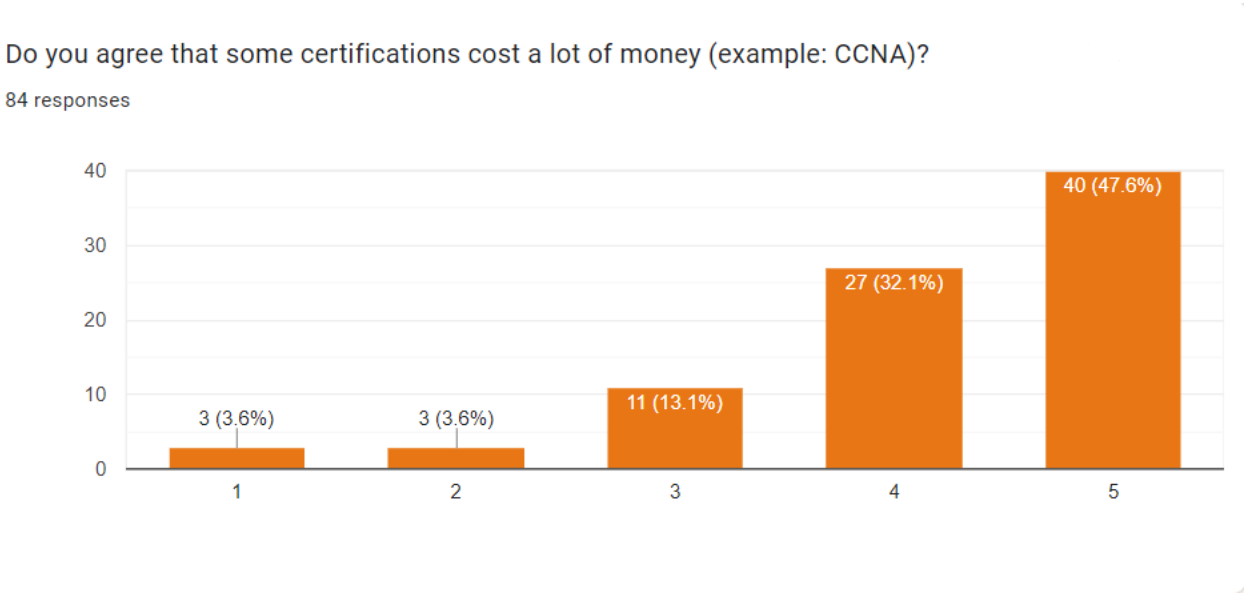

Figure 7 - Questionnaire Question 6

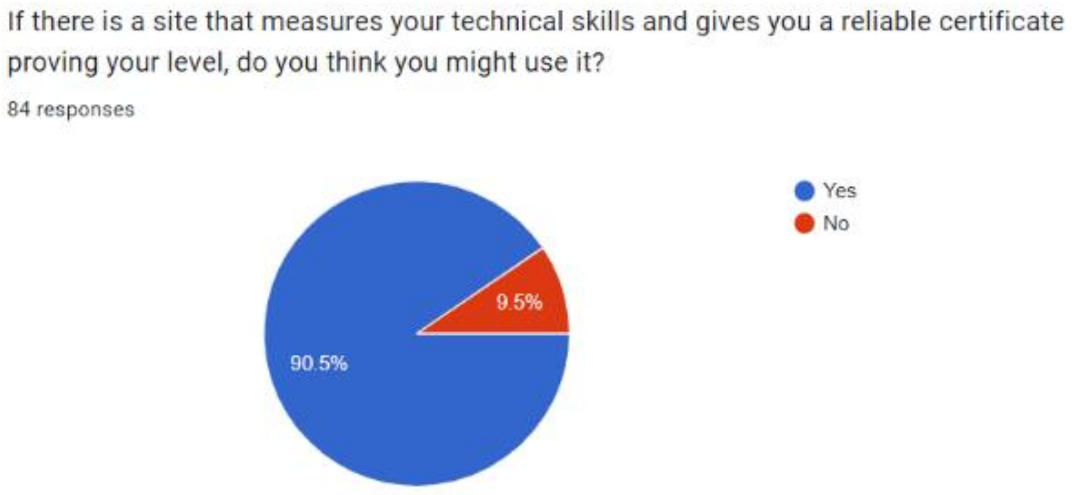

Figure 8 - Questionnaire Question 7

3.3.2 Interview

We met with people in need of this system (students), and we asked them these questions:

- Are you interested in measuring your technical skills and obtaining an accredited certificate that proves your level?
- Have you ever studied a course on one of the sites that provide courses "Udemy - Coursera - etc."?
- Do you agree that some certifications cost a lot of money (example: CCNA)?
- If there is a site that measures your technical skills and gives you a reliable certificate proving your level, do you think you might use it?

We learned from the responses that students require such a system to benefit from it in proving their skills and abilities, particularly to employers, for free, without having to pay large sums of money for exams and from the comfort of their own homes in a modern and simple new method.

3.4 Targeted Users

The targeted users of the Mo3tamad website are people who want to improve their skills and want an authorized certificate to prove their abilities.

3.5 Functional Requirements Definition

User:

- Signup
- Login
- View available exam.
- Select exam.
- Strat exam.
- End exam.
- View result.
- Download certificate.
- Logout.
- Write comment.
- Edit comment.
- Delete comment.
- Rating exam.
- Edit profile.

Admin (company):

- Login.
- Signup.
- Create exam.
- Edit exam.
- Delete exam.
- Update exam
- Create question.
- Edit question.
- Delete question.
- Update question.
- Create assistant user.
- Remove assistant user.
- Applicant approves.
- Applicant declines.
- Mange company profile.

Assistants:-

- Login.
- Create exams.
- Delete exam.
- Update exam
- Create question.
- Edit question.
- Delete question.

- Update question
- Applicant approves.
- Applicant declines.
- Edit profile.

System:

- Generate certificate.
- Generate exam.
- Calculate the result of user.
- Send email to user contain exam result.
- Switch into full screen to start exam.
- End exam if the user cheated.
- End exam if the user exceed the time limit.

3.6 Functional Requirements Specification

Requirement No.	Requirement name	Description
R1	Signup (User)	• Create account from sign up page • Fill form with data: first name, last name, email, password, age, gender, phone number.
R2	Login (User)	Enter email and password in the specified field
R3	View available exam (User)	Users can discover available exams on the website.
R4	Select exam	user attend exam after selecting one
R5	Strat exam	User can start exam at the exam time range.
R6	End exam	User can end the exam by end button if the time doesn't exceed.
R7	View result	user view result after finishing the exam
R8	Download certificate	download certificate after system approve the result

R9	Write comment	User can write comment about the exam after he finished it.
R10	Edit comment	User can edit the comment
R11	Delete comment	User can delete his comment on the certain exam.
R12	Rating exam	User can rate the exam after he finished it .
R13	Logout (User)	after user's login into site they, can select logout option to logout form the site
R14	Sign up(company)	Admin can sign up as a company to have more privileges than other users.
R15	Login (company)	Enter email and password that the company sign up with.
R16	Create exam	create exam template that contains the questions
R17	Delete exam	Admin can delete exam that he has made before .
R18	Update exam	Admin can update exam features.
R19	Create question	Admin can create question in a specific exam
R20	Edit question	Admin may edit the question that he has made before
R22	Delete question	Admin can delete the question he has made before.
R23	Create assistant users	Admin can make assistant account to help him with some duties.
R24	Remove assistant user	Admin can delete the assistant account that he made.
R25	Applicant approves	After user attend and finish exam, admin can approve it.
R26	Applicant declines	If user cheated admin will decline the applicant
R27	Mange company profile	Edit profile information

R28	Logout	Admin can logout form the system
R29	Login (assistant)	Assistant will login by the information form the admin
R30	Create exam	create exam template that contains the questions
R31	Delete exam	Admin can delete exam that he has made before .
R32	Update exam	Admin can update exam features.
R33	Create question	Admin can create question in a specific exam
R34	Edit question	Admin may edit the question that he has made before
R35	Delete question	Admin can delete the question he has made before.
R36	Applicant approves	After user attend and finish exam, assistant can approve it.
R37	Applicant declines	If user cheated assistant can decline the applicant
R38	Edit profile	Assistant can edit his profile
R39	Generate certificate	After user passed the exam and admin approve it system will generate certificate
R40	Generate exam	System will generate exam with it's features
R41	Calculate the result	System should calculate user result when he finish the exam
R42	Send email to users	System will send emails to users when they enroll to exam
R43	Switch into full screen mode	User can't start the exam if he doesn't enable full screen mode

R44	End exam when cheated	If user cheated the system will end the exam
R45	End exam if the user exceeds the time limit	If user exceeds the time of exam the system will end the exam

Table 7 - Functional Requirements Specification

3.7 Non-functional requirements

Requirement No.	Requirement	Specification
1	Usability	The website should be efficient to use, easy to use and has a consistent interface.
2	Security	Emails & passwords for admins and users must be kept secure, to prevent tampering.
3	Robustness	The site must bear the maximum number of errors and malfunctions without fail
4	Availability	The website should be able to operate 24/7 because users will attend to exams in different time.
5	Performance	The website must be efficient and provide all these functions in very quick time with suitable algorithms.
6	Maintainability	The website shall be easy to maintain; faults must be easy to correct
7	Modifiability	The site must be easy, quick, and cheap to respond to changing & increasing needs

Table 8 - Non-functional requirements

3.8 Summary

Finally, this chapter defines our system requirements and their analysis, which includes a detailed description of the system's functionality, plus the specification for each requirement, it also includes the non-functional requirements that support our system.

4.0 CHAPTER FOUR: SYSTEM DESIGN

4.1 Introduction

Modeling Diagrams help to clarify and communicate ideas about the user requirements that the software system must support. As well as they identify and describe functional behavior and structure of the software. In this chapter we will focus on system design, the context diagram that clarifies the system requirements and their relationships are addressed in section 4.2, Data flow diagrams DFD in section 4.3, in section 4.4 is Entity Relationship Diagram, UML Use Case diagram in section 4.5, UML Sequence diagram in section 4.6, UML Class diagram in 4.7 & Graphical User Interface (GUI) Design in 4.8.

4.2 Context Diagram

Context diagram: sometimes called a level 0 data-flow diagram, is drawn to define and clarify the boundaries of the software system. It identifies the flows of information between the system and external entities.

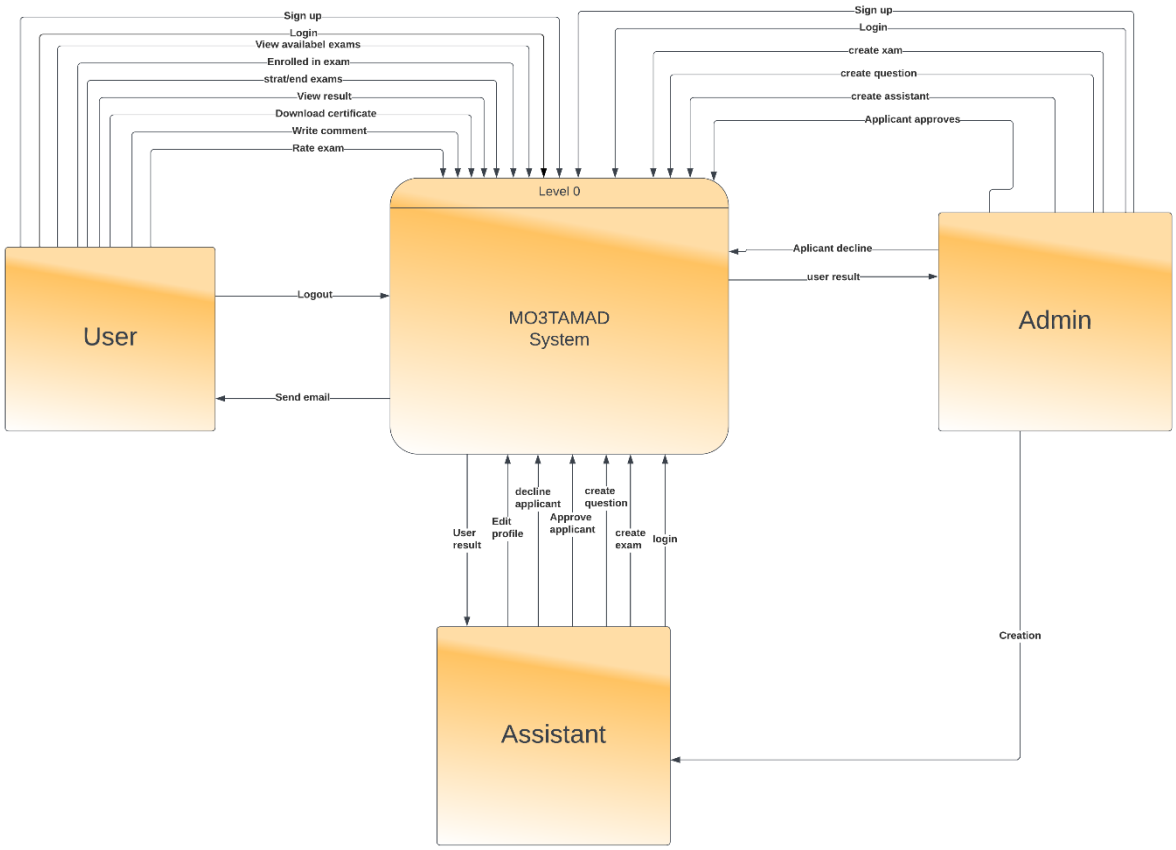

Figure 9 - Context Diagram

4.3 Data Flow Diagram (DFD)

Known as DFD, Data flow diagrams are used to graphically represent the flow of data in a business information system. DFD describes the processes that are involved in a system to transfer data from the input to the file storage and reports generation.

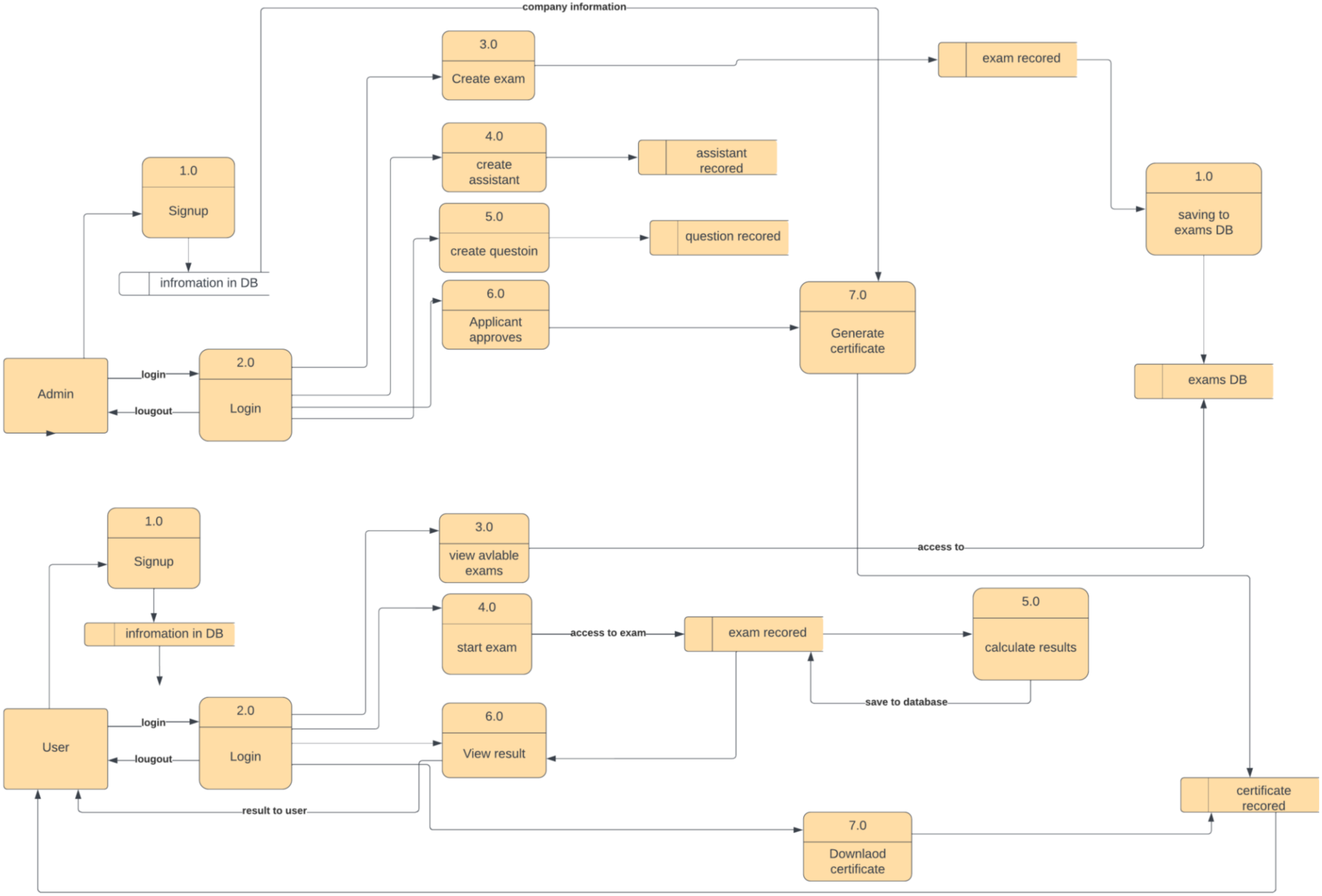
company information
3.0
Create exam
exam recored
4.0
create assistant
assistant recored
1.0
Signup
1.0
saving to exams DB
5.0
create questoin
question recored
infromation in DB
6.0
Applicant approves
7.0
Generate certificate
Admin
login
lougout
2.0
Login
exams DB
1.0
Signup
3.0
view avlable exams
access to
infromation in DB
4.0
start exam
access to exam
exam recored
5.0
calculate results
save to database
User
login
lougout
2.0
Login
6.0
View result
certificate recored
result to user
7.0
Downlaod certificate

Figure 10 - Data Flow Diagram (DFD)

4.5 UML Use Case Diagram

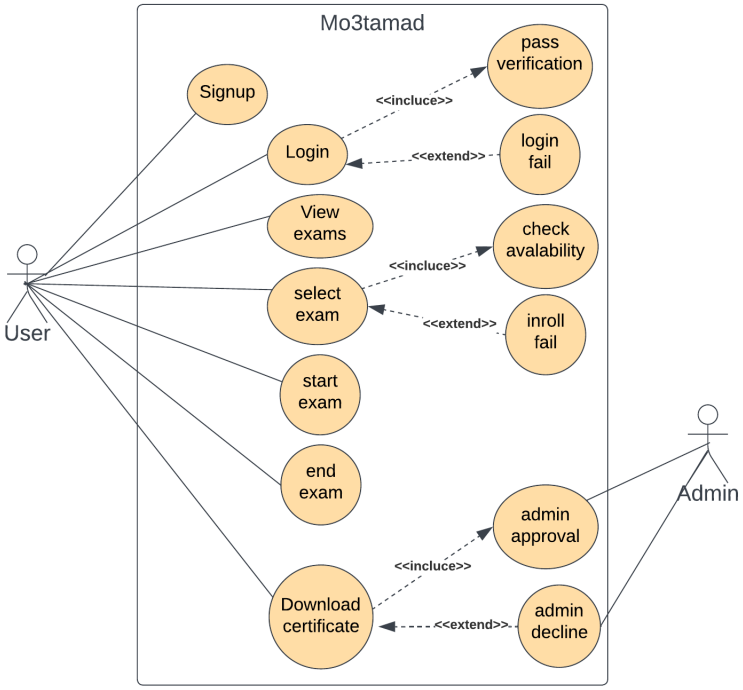

Figure 11 - UML Use Case Diagram 1

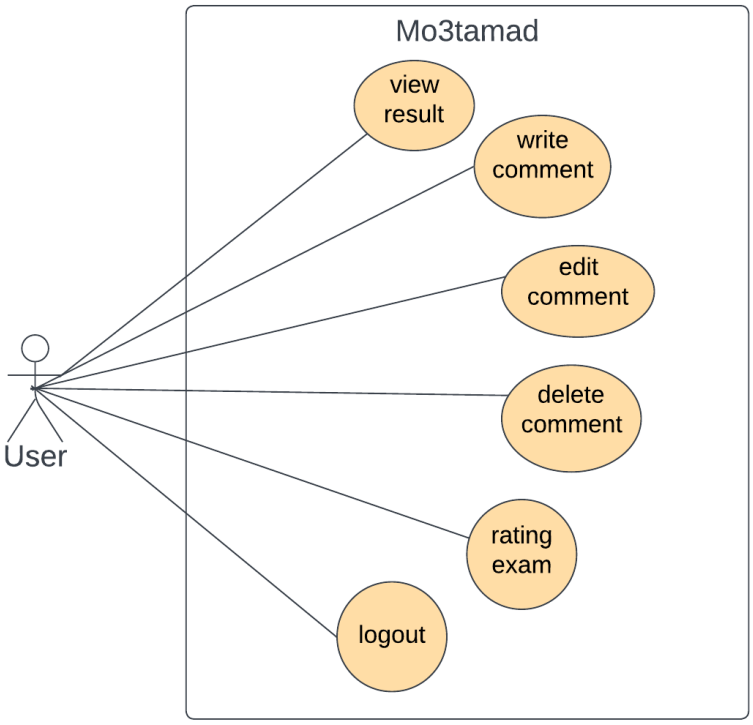

Figure 12 - UML Use Case Diagram 2

Figure 13-UML Use case diagram 3

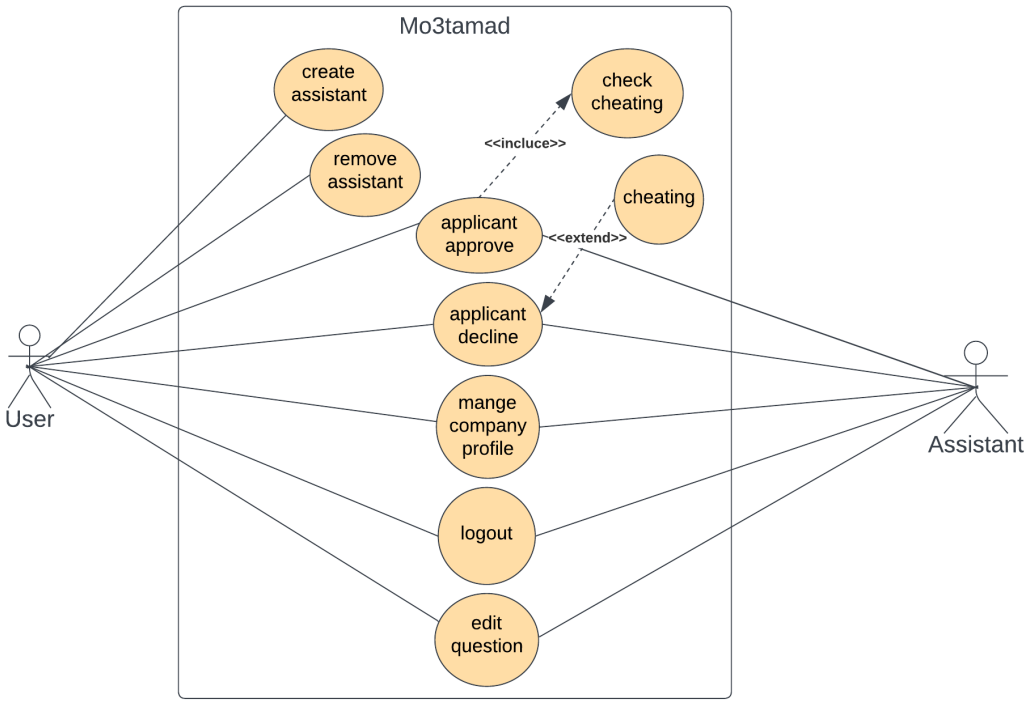

Figure 14- UML Use case diagram 4

4.6 UML Sequence Diagram

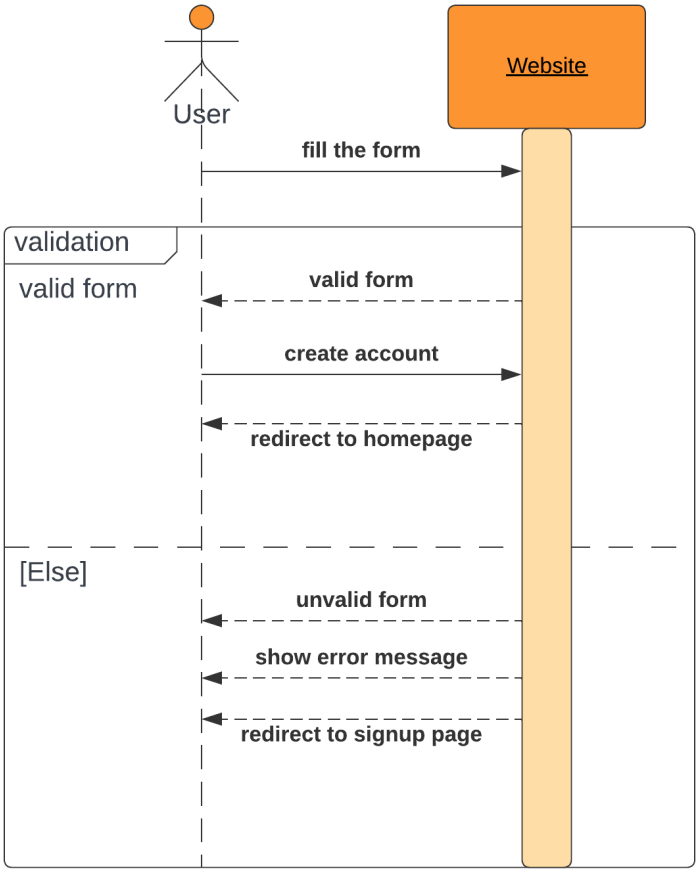

Figure 15 - UML Sequence Diagram 1

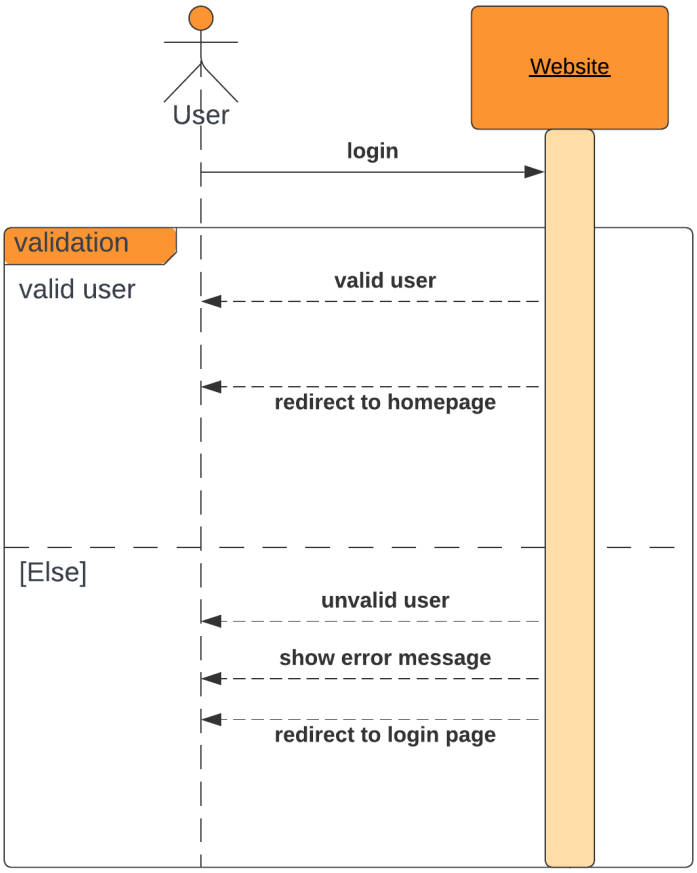

Figure 16 - UML Sequence Diagram 2

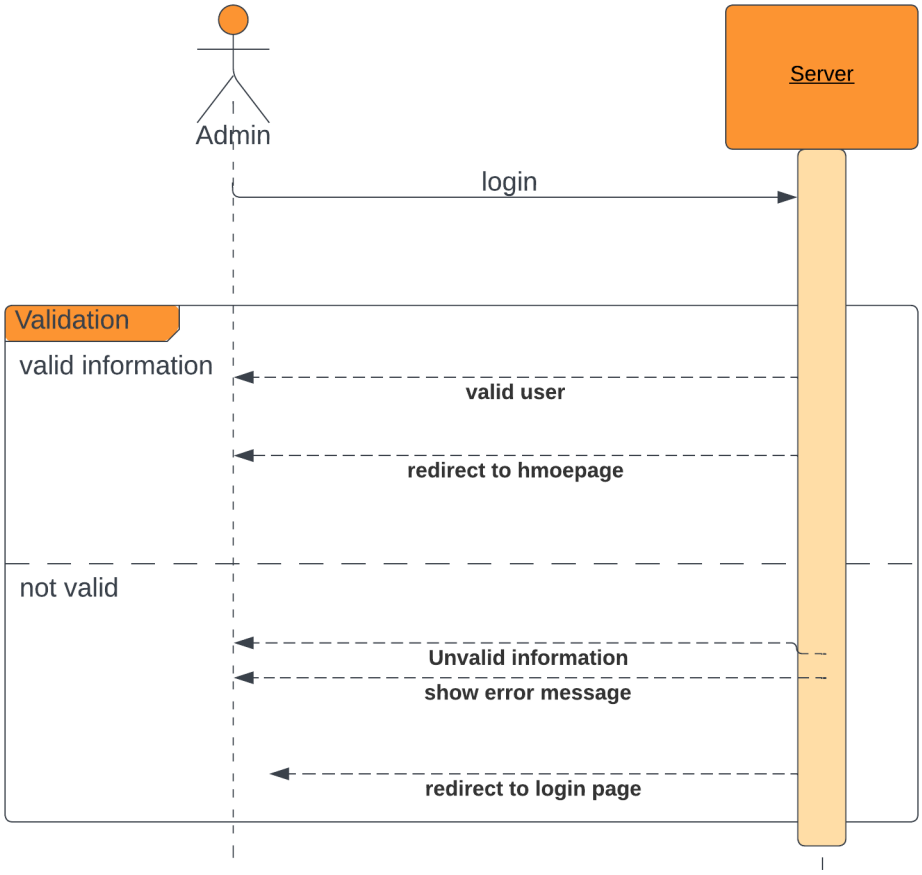

Figure 17UML_Sequence3

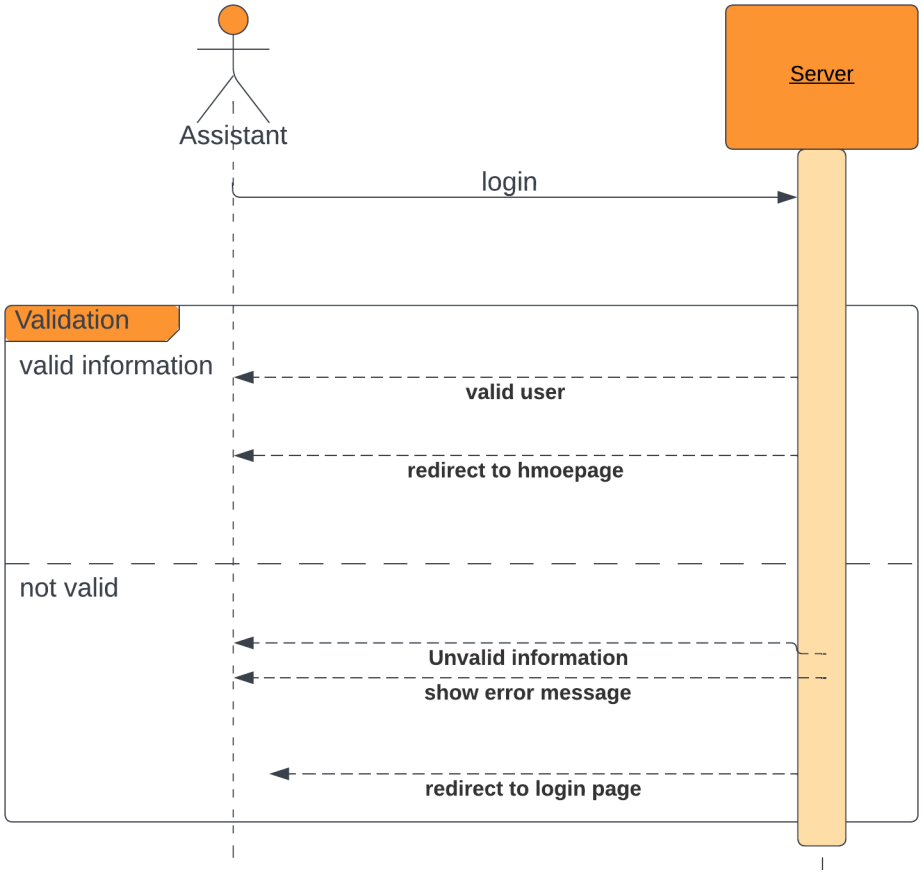

Figure 18UML sequence_4

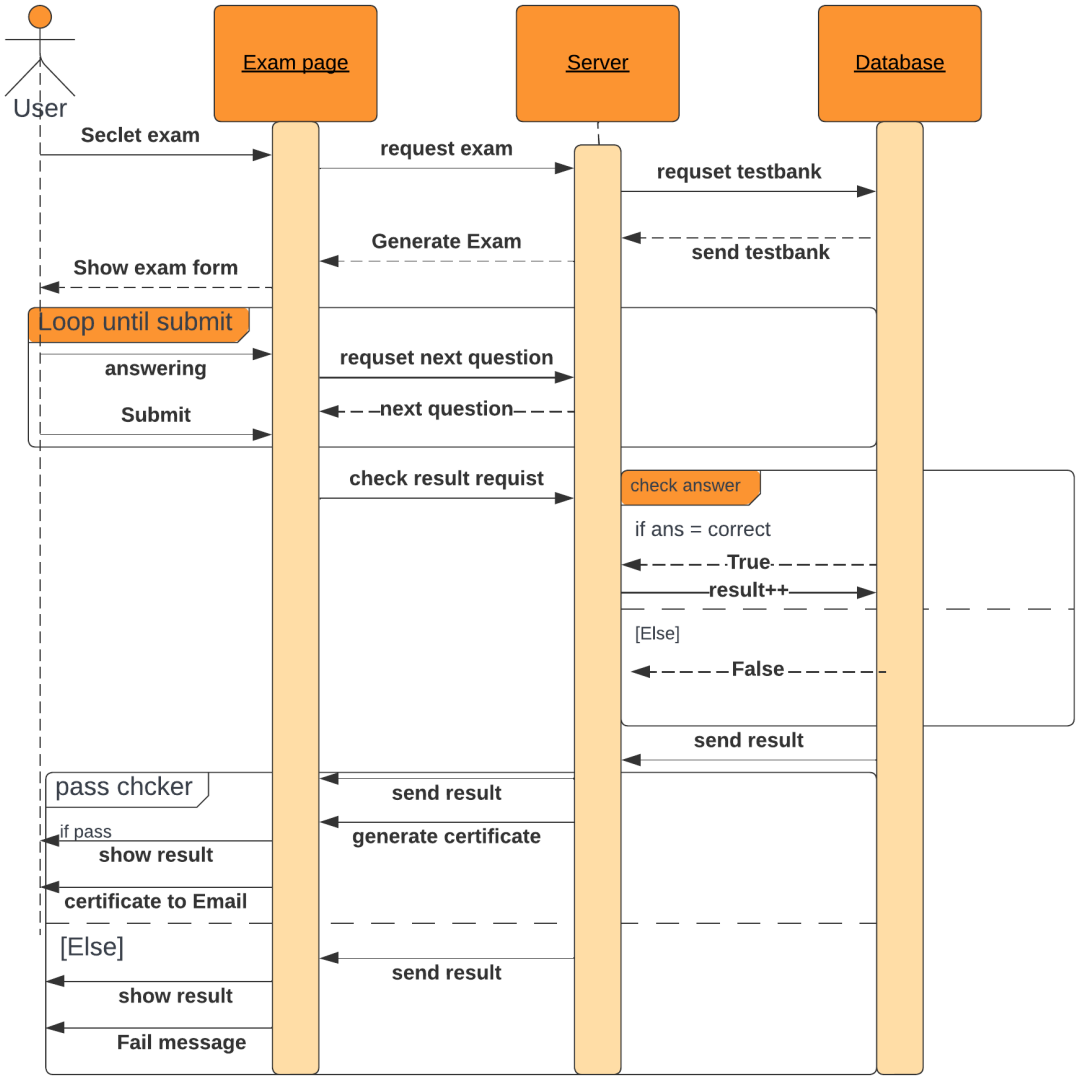

Figure 19 - UML Sequence Diagram_5

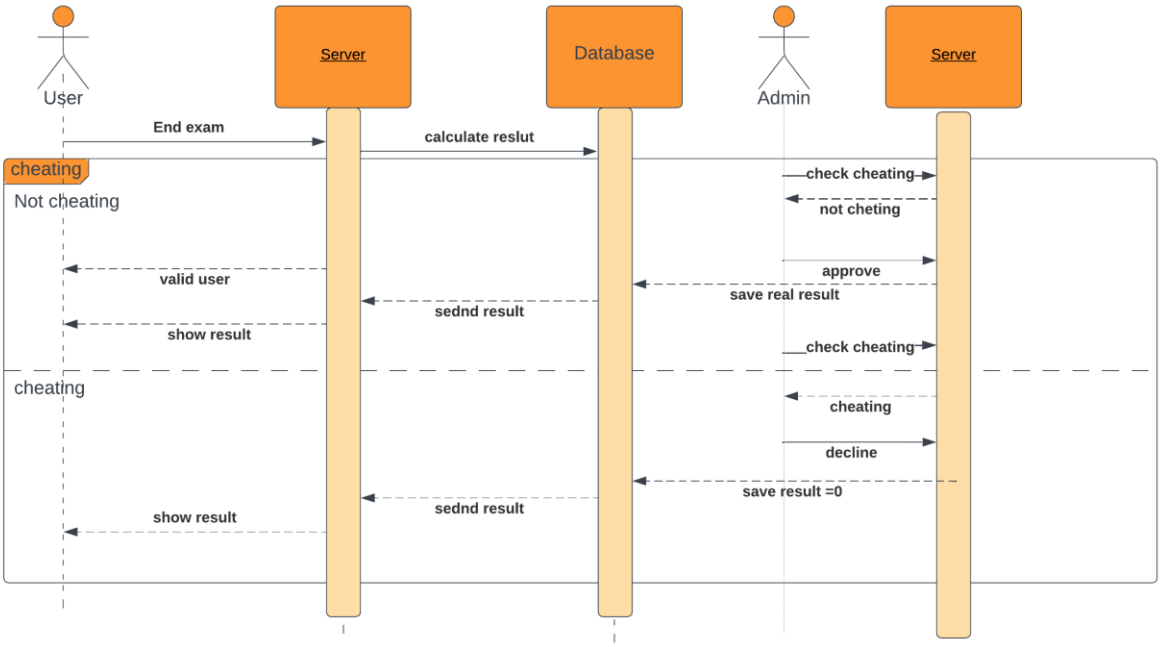

Figure 20-UML-Sequence_6

4.7 UML Class Diagram

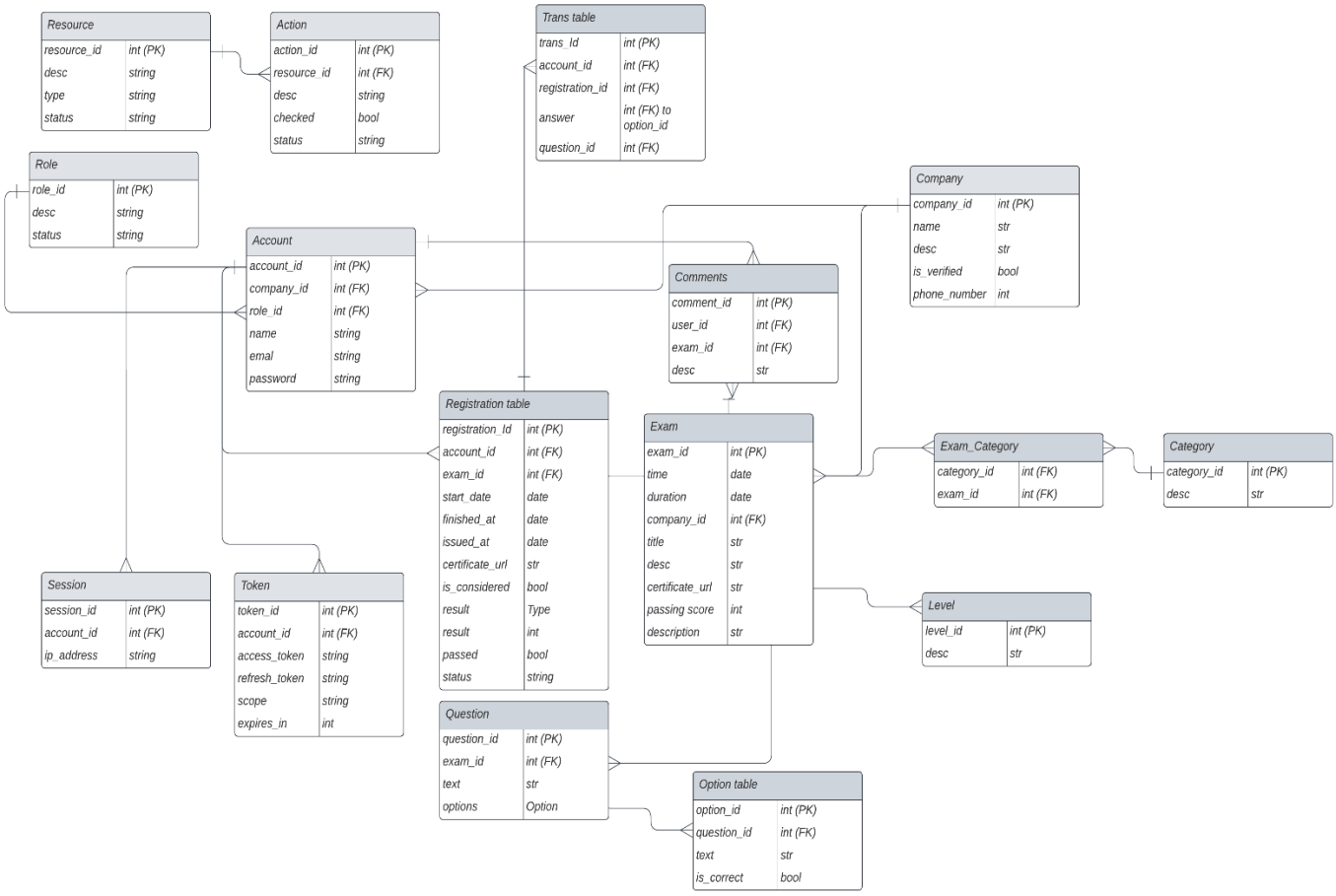

Figure 21 - UML Class Diagram

4.8 Graphical User Interface (GUI) Design

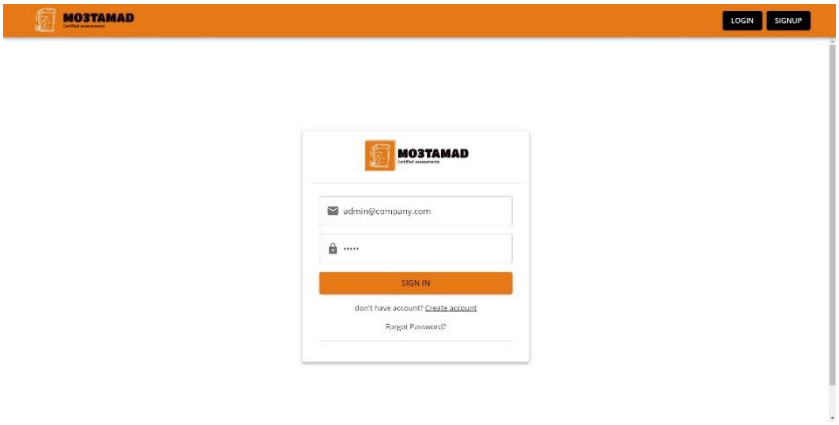

Figure 22 – login page

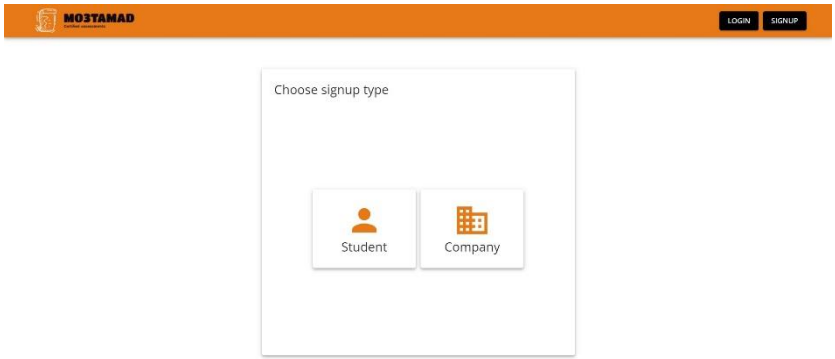

Figure 23 – SignUp Page

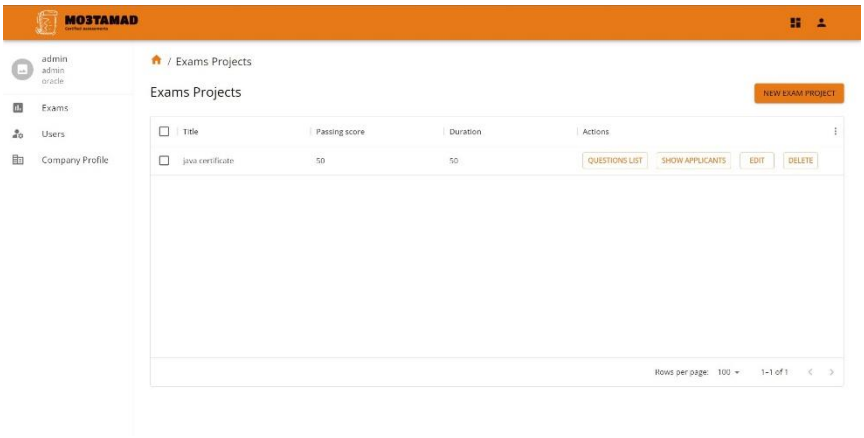

Figure 24 – Dashboard

Figure 25 - Exams Page

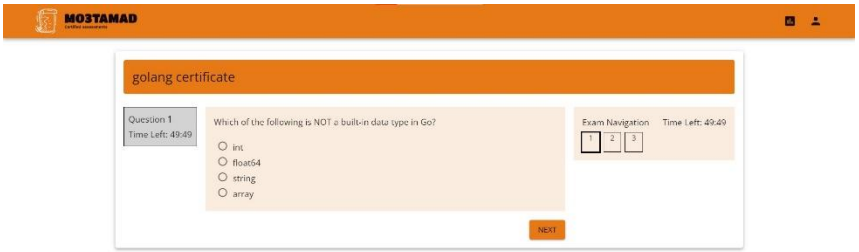

Figure 26 – Exam Start

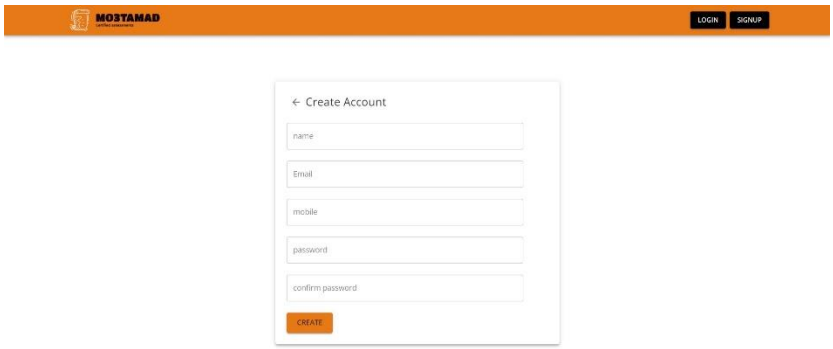

Figure 27 – create account

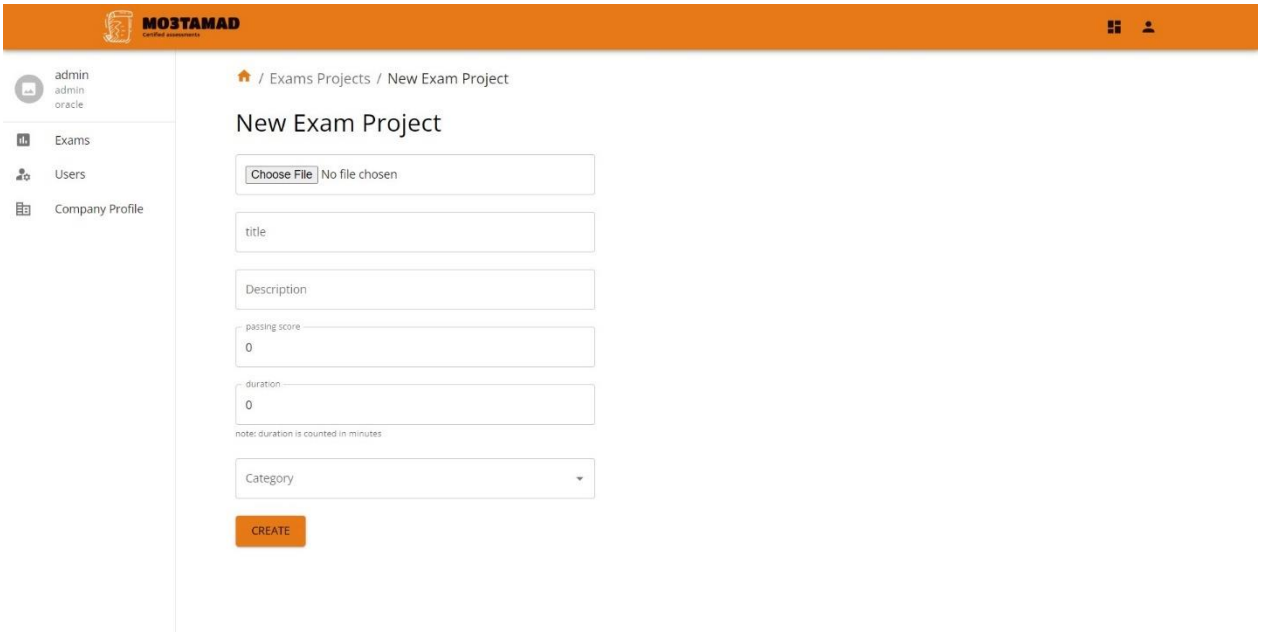

Figure 28-CreateExam

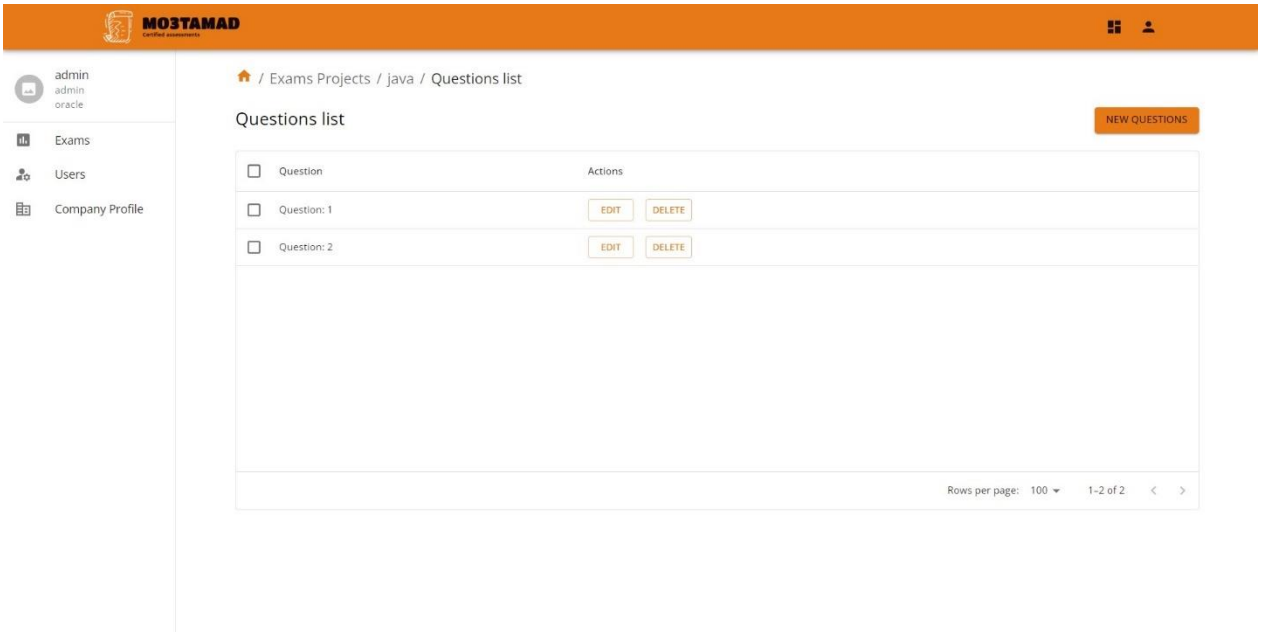

Figure 29_Question List

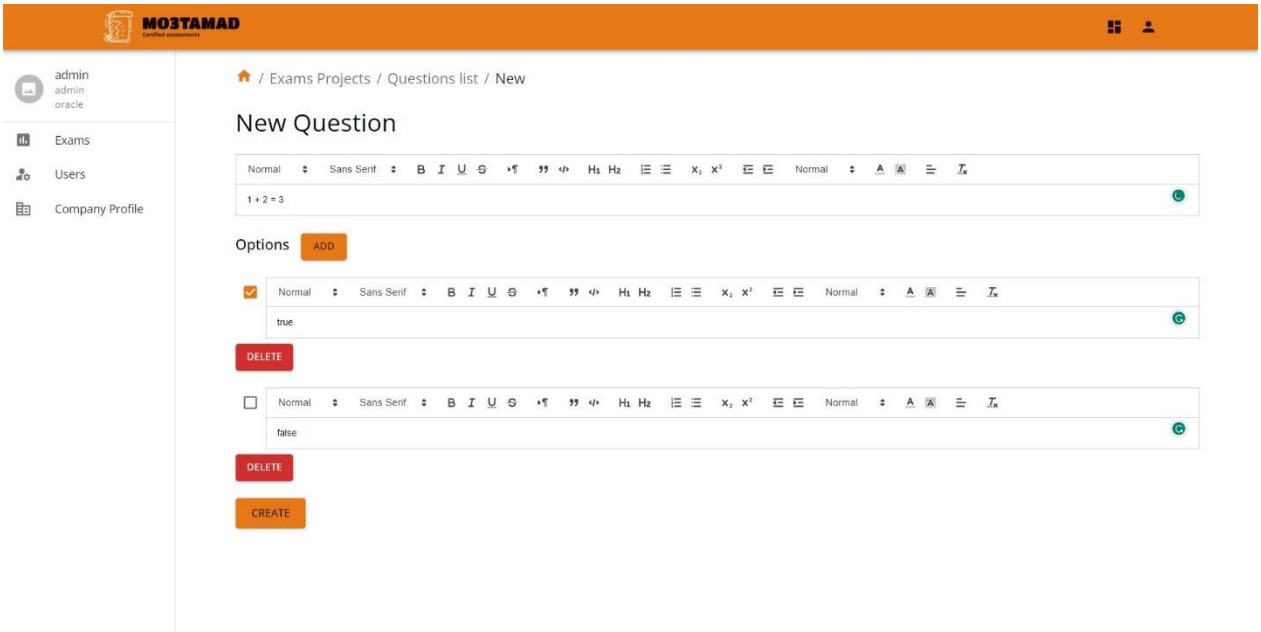

Figure 30-New-Question

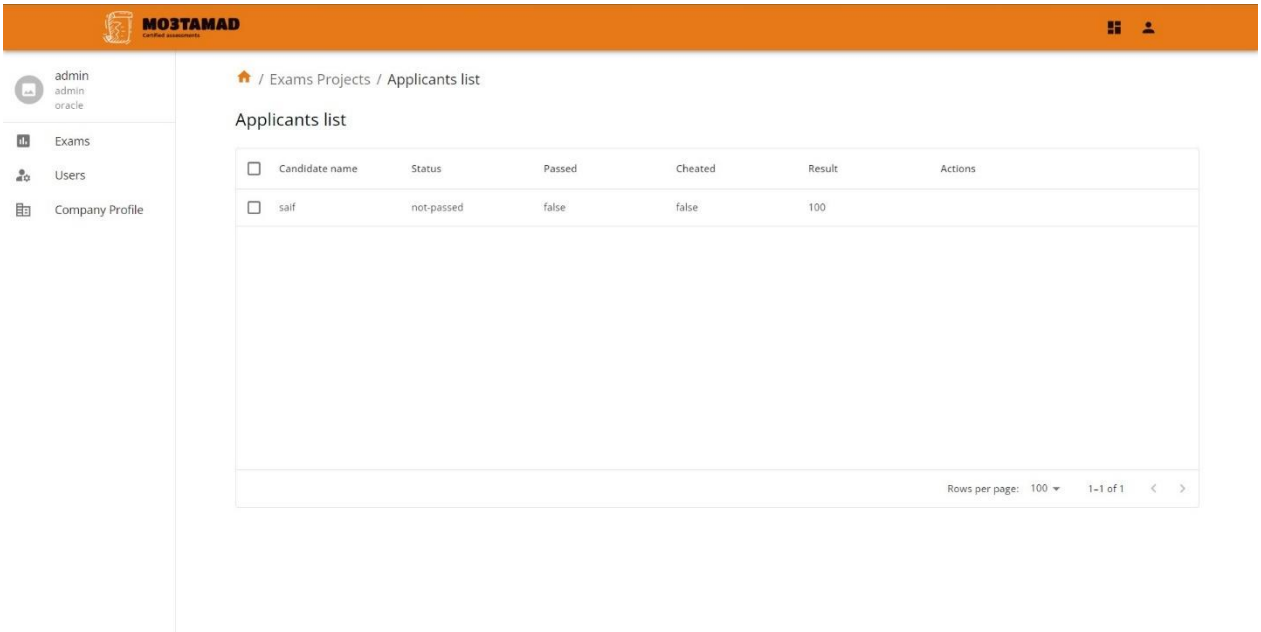

Figure 31-Applicants-list

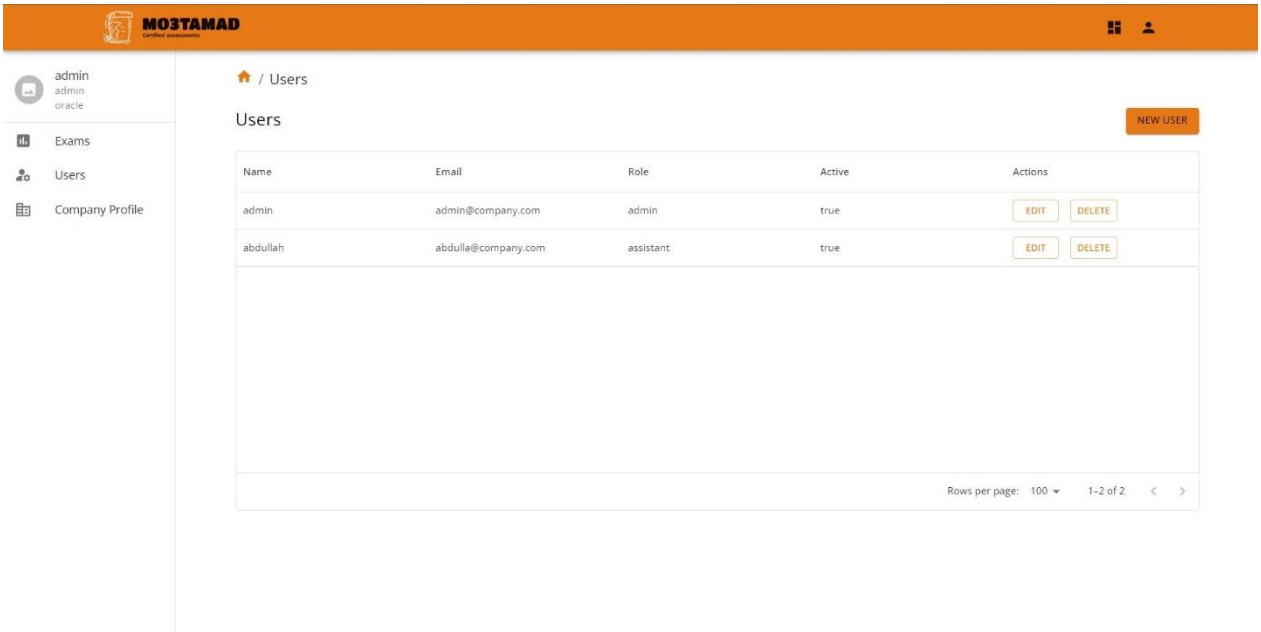

Figure 32-Users

Figure 33-Company-Profile

4.9 Summary

In this chapter, we illustrated the Context-diagram, Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), UML use case diagram, sequence diagram, and a class diagram, and we made the graphical user interface for our website.

5.0 System implementation

5.1 Introduction :-

In this chapter, the website implementation is discussed. We will talk about implementation tools and database implementation.

5.2 Database implementation

We are using a relational database, specifically MySQL, to store data related to registrations, accounts, exams, questions, company, option, and transactions. The database models, defined as Go structs, represent the tables in the database, and the GORM library is utilized to map the struct fields to the corresponding database columns.

To interact with the database, our Go application executes SQL queries or uses an ORM (Object-Relational Mapping) library like GORM to perform common database operations such as creating, reading, updating, and deleting records (CRUD operations).

Code Blame Executable File · 18 lines (17 loc) · 1.07 KB Raw

```go
package model

type Account struct {
        AccountId     int           `gorm:"primaryKey;autoIncrement:true;column:account_id" json:"id,omitempty"`
        Name          string        `gorm:"column:name" json:"name,omitempty"`
        Email         string        `gorm:"unique;column:email" json:"email,omitempty"`
        Mobile        string        `gorm:"column:mobile" json:"mobile,omitempty"`
        Password      string        `gorm:"column:password" json:"password,omitempty"`
        RoleID        int           `gorm:"column:role_id" json:"roleId,omitempty"`
        Role          Role          `gorm:"foreignKey:RoleID" json:"role,omitempty"`
        Active        bool          `gorm:"default:true;column:active" json:"active"`
        Status        string        `gorm:"column:status" json:"status,omitempty"`
        CompanyId     int           `gorm:"column:company_id" json:"companyId,omitempty"`
        Company       Company       `json:"company,omitempty"`
        Registrations []Registration `gorm:"forigenKey:AccountId" json:"registrations,omitempty"`
        Comments      []Comment      `gorm:"forigenKey:AccountId" json:"comments,omitempty"`
        Model
}
```

Figure 34-Account-Struct

Code Blame Executable File · 8 lines (7 loc) · 311 Bytes Raw

```go
package model

type Action struct {
        ActionId   int    `gorm:"primaryKey;autoIncrement:true;column:id" json:"id"`
        Desc       string `gorm:"column:desc" json:"desc,omitempty"`
        ResourceId int    `gorm:"column:resource_id" json:"resourceId,omitempty"`
        Checked    bool   `gorm:"column:checked" json:"checked"`
}
```

Figure 35-Action-Struct

Code Blame 11 lines (10 loc) · 477 Bytes Raw

```go
package model

type Comment struct {
	CommentId int     `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	ExamId    int     `gorm:"column:exam_id" json:"examId,omitempty"`
	AccountId int     `gorm:"column:accountId" json:"accountId,omitempty"`
	Desc      string  `gorm:"column:desc" json:"desc,omitempty"`
	Exam      Exam    `gorm:"forigenKey:ExamId" json:"exam,omitempty"`
	Account   Account `gorm:"forigenKey:AccountId" json:"account,omitempty"`
	Model
}
```

Figure 36-Comment-struct

Code Blame 11 lines (10 loc) · 497 Bytes Raw

```go
package model

type Company struct {
	CompanyId    int    `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	Name         string `gorm:"column:name" json:"name,omitempty"`
	Desc         string `gorm:"column:desc" json:"desc,omitempty"`
	IsVerified   bool   `gorm:"default:true;column:is_verified" json:"isVerified"`
	Phone_number string `gorm:"column:phone_number" json:"phoneNumber,omitempty"`
	Exams        []Exam `gorm:"forigenKey:CompanyId" json:"exams,omitempty"`
	Model
}
```

Figure 37-company-s

Code Blame 23 lines (21 loc) · 1.42 KB Raw

```go
package model

import "time"

type Exam struct {
	ExamID           int            `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	Duration         time.Duration  `gorm:"column:duration" json:"duration,omitempty"`
	Title            string         `gorm:"column:title" form:"name" json:"name,omitempty"`
	Description      string         `gorm:"column:escription" form:"desc" json:"desc,omitempty"`
	ThumbnailUrl     string         `gorm:"column:thumbnail_url" json:"thumbnailUrl,omitempty"`
	CertificationUrl string         `gorm:"column:certification_url" json:"certificationUrl,omitempty"`
	PassingScore     int            `gorm:"column:passing_score" json:"passingScore,omitempty"`
	CompanyId        int            `gorm:"column:company_id" json:"companyId,omitempty"`
	Company          Company        `gorm:"forigenKey:CompanyId" json:"company,omitempty"`
	QuestionsCount   int            `gorm:"column:questions_count" json:"questionsCount"`
	LevelId          int            `gorm:"column:level_id" json:"levelId,omitempty"`
	Level            Level          `gorm:"forigenKey:LevelId" json:"level"`
	Questions        []Question     `gorm:"forigenKey:ExamId" json:"questions"`
	Registrations    []Registration `gorm:"forigenKey:ExamId" json:"registrations,omitempty"`
	Comments         []Comment      `gorm:"forigenKey:ExamId" json:"comments,omitempty"`
	Categories       []*Category    `gorm:"many2many:exam_categories;"`
	Model
}
```

Figure 38-Exam-Struct

Code Blame 10 lines (9 loc) · 435 Bytes Raw

```go
package model

type Option struct {
	OptionId   int       `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	QuestionId int       `gorm:"column:question_id" json:"questionId,omitempty"`
	Text       string    `gorm:"column:text" json:"text,omitempty"`
	IsCorrect  bool      `gorm:"column:is_correct" json:"isCorrect,omitempty"`
	Question   *Question `gorm:"forigenKey:question_id" json:"question,omitempty"`
	Model
}
```

Figure 39-Option-Struct

Code Blame 10 lines (9 loc) · 435 Bytes Raw

```go
package model

type Question struct {
	QuestionId int      `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	ExamId     int      `gorm:"column:exam_id" json:"examId,omitempty"`
	Text       string   `gorm:"column:text" json:"text,omitempty"`
	Options    []Option `gorm:"foreignKey:QuestionId;OnDelete:CASCADE;" json:"options,omitempty"`
	Exam       *Exam    `gorm:"forigenKey:ExamId" json:"exam,omitempty"`
	Model
}
```

Figure 40-Question-Struct

Code Blame 23 lines (21 loc) · 1.24 KB Raw

```go
package model

import "time"

type Registration struct {
	RegistrationId int        `gorm:"primaryKey;autoIncrement:true;column:id" json:"id,omitempty"`
	AccountId      int        `gorm:"column:account_id" json:"accountId,omitempty"`
	ExamId         int        `gorm:"column:exam_id" json:"examId,omitempty"`
	StartTime      *time.Time `gorm:"column:start_time" json:"startTime"`
	EndTime        *time.Time `gorm:"column:end_time" json:"endTime"`
	FinishedAt     *time.Time `gorm:"column:finished_at" json:"finishedAt"`
	IsCheated      bool       `gorm:"column:is_cheated" json:"isCheated"`
	IsConsidered   bool       `gorm:"column:is_considered" json:"isConsidered"`
	CertificateUrl string     `gorm:"column:certificate_url" json:"certificateUrl"`
	IssuedAt       *time.Time `gorm:"column:issued_at" json:"issuedAt"`
	Passed         bool       `gorm:"column:passed" json:"passed"`
	Result         int        `gorm:"column:result" json:"result"`
	Status         string     `gorm:"column:status" json:"status,omitempty"`
	Account        Account    `gorm:"foriegnKey:AccountId" json:"account,omitempty"`
	Exam           Exam       `gorm:"forigenKey:ExamId" json:"exam,omitempty"`
	Trans          []Trans    `gorm:"forigenKey:RegistrationId" json:"trans,omitempty"`
	Model
}
```

Figure 41-Registration-Struct

5.3 Other components Implementation

- **API Routes**: In our project, we have defined various API routes using a Golang web framework called Fiber.
  API routes define the endpoints that clients can access to interact with our application. Each route is associated with a specific HTTP method (e.g., POST, GET, DELETE) and handles incoming requests by executing the corresponding handler function. These routes define the entry points for different operations such as registration, starting/finishing exams, retrieving data, and deleting registrations.

```go
func (s *Server) Register() {
        //************************ Route Grouping ******************************
        api := s.App.Group("/api")
        v1 := api.Group("/v1")

        //************************ Middlewares ******************************
        s.App.Use(cors.New())
        s.App.Use(s.Middleware.RequestsLogger)
        api.Use(s.Middleware.Protect)

        //************************ AUTH Routes ******************************
        oauthRoutes := s.App.Group("auth/oauth")
        oauthRoutes.Post("/login", s.Middleware.BasicAuthParser, s.Login)
        oauthRoutes.Post("/signup", s.Signup)
        oauthRoutes.Post("/signup/company", s.SignupCompany)
        oauthRoutes.Delete("/logout/:session_id", s.Logout)
        oauthRoutes.Post("/refresh/token", s.RefreshToken)
```

Figure 42-Route-1

```
//************************ System Routes ****************************
system := v1.Group("/system")

// protect all system routes
system.Use(s.Middleware.Authorization(authz.Admin))

// Roles
roleRoutes := system.Group("/roles")
roleRoutes.Get("/", s.GetAllRoles)
roleRoutes.Post("/", s.CreateRole)
roleRoutes.Put("/:role_id", s.UpdateRole)
roleRoutes.Delete("/:role_id", s.DeleteRole)

// Resources
resourceRoutes := system.Group("/resources")
resourceRoutes.Get("/", s.GetAllResources)
resourceRoutes.Get("/:role_id/role", s.GetAllResourcesWithRole)
resourceRoutes.Post("/", s.CreateResource)
resourceRoutes.Put("/:resource_id", s.UpdateResource)
resourceRoutes.Delete("/:resource_id", s.DeleteResource)

// Policies
policyRoutes := system.Group("/policies")
policyRoutes.Get("/", s.GetAllPolicies)
policyRoutes.Get("/:role", s.GetRolePolicies)
policyRoutes.Post("/", s.CreatePolicies)
policyRoutes.Put("/", s.UpdatePolicy)
policyRoutes.Delete("/", s.DeletePolicies)

//************************ Business Routes ****************************
```

Figure 43-Route-2

```
// Accounts
accountsRoutes := v1.Group("/accounts")
accountsRoutes.Get("/", s.Middleware.Authorization(authz.Admin), s.GetAllAccounts)
accountsRoutes.Get("/company/:companyId", s.Middleware.Authorization(authz.GetUsers), s.GetAllAccountsByCompanyId
accountsRoutes.Get("/me", s.GetMyProfile)
accountsRoutes.Patch("/me", s.UpdateMyProfile)
// this is an exeption because we need every account to be able to access it
accountsRoutes.Get("/policies/ui", s.GetAllPoliciesRoleUI)
accountsRoutes.Get("/:id", s.Middleware.Authorization(authz.GetUser), s.GetAccount)
accountsRoutes.Post("/", s.Middleware.Authorization(authz.CreateUser), s.CreateAccount)
accountsRoutes.Delete("/:id", s.Middleware.Authorization(authz.DeleteUser), s.DeleteAccount)
accountsRoutes.Patch("/:id", s.Middleware.Authorization(authz.UpdateUser), s.UpdateAccount)

// Companies
//Exams
examsRoutes := v1.Group("/exams")
examsRoutes.Get("/", s.GetAllExams)
examsRoutes.Get("/company/:companyId", s.Middleware.Authorization(authz.GetExams), s.GetAllExamsByCompanyId)
examsRoutes.Get("/:id", s.GetExam)
examsRoutes.Post("/", s.Middleware.Authorization(authz.CreateExam), s.CreateExam)
examsRoutes.Delete("/:id", s.Middleware.Authorization(authz.DeleteExam), s.DeleteExam)
examsRoutes.Patch("/:id", s.Middleware.Authorization(authz.UpdateExam), s.UpdateExam)

questionRoutes := v1.Group("/questions")
questionRoutes.Get("/", s.GetAllQuestions)
questionRoutes.Get("/:id", s.GetQuestion)
questionRoutes.Get("/byId/:id", s.GetQuestionByExamId)
questionRoutes.Post("/", s.CreateQuestion)
questionRoutes.Delete("/:id", s.DeleteQuestion)
questionRoutes.Patch("/:id", s.UpdateQuestion)
```

Figure 44-Route-3

```
registrationRoutes := v1.Group("/registration")
registrationRoutes.Get("/my-exams", s.GetRegByUserId)
registrationRoutes.Get("/:id", s.GetRegistration)
registrationRoutes.Get("/byExam/:id", s.GetRegByExamId)
registrationRoutes.Delete("/:id", s.DeleteRegistration)
registrationRoutes.Patch("/:id/start", s.Middleware.Authorization(authz.Register), s.StartedExam)
registrationRoutes.Patch("/:id/approve", s.Middleware.Authorization(authz.CreateExam), s.ApproveApplicant)
registrationRoutes.Patch("/:id/decline", s.Middleware.Authorization(authz.CreateExam), s.DeclineApplicant)
registrationRoutes.Patch("/:id/finish", s.Middleware.Authorization(authz.Register), s.FinishedExam)
registrationRoutes.Patch("/:id/cheat", s.Middleware.Authorization(authz.Register), s.UserCheated)
registrationRoutes.Post("/:account_id/:exam_id", s.Middleware.Authorization(authz.Register), s.CreateRegis

companyRoutes := v1.Group("/company")
companyRoutes.Get("/", s.GetAllCompanies)
companyRoutes.Get("/:id", s.GetCompany)
companyRoutes.Post("/", s.CreateCompany)
companyRoutes.Delete("/:id", s.DeleteCompany)
companyRoutes.Patch("/:id", s.UpdateCompany)

transRoutes := v1.Group("/trans")
transRoutes.Post("/:transId/answer", s.UserAnswerd)
//transRoutes.Get("/:id",s.GetUserAnswers)

// thumbnails
s.App.Get("/thumbnails/:thumbnailUrl", s.GetThumbnail)
s.App.Get("/certificates/:certUrl", s.GetCertificate)
```

Figure 45-Route-4

```go
        // thumbnails
        s.App.Get("/thumbnails/:thumbnailUrl", s.GetThumbnail)
        s.App.Get("/certificates/:certUrl", s.GetCertificate)

        // comments
        commentRoutes := v1.Group("/comment")
        commentRoutes.Get("/", s.GetAllComments)
        commentRoutes.Post("/", s.CreateComment)
        commentRoutes.Patch("/:id", s.UpdateComment)
        commentRoutes.Delete("/:id", s.DeleteComment)

        // categories
        categoryRoutes := v1.Group("/category")
        categoryRoutes.Get("/", s.GetAllCategories)

        // levels
        levelRoutes := v1.Group("/level")
        levelRoutes.Get("/", s.GetAllLevels)

}
```

Figure 46-Route-5

- **API**: our API endpoints represent the different URLs that clients can access to interact with our application. Each endpoint corresponds to a specific operation or resource in your application. For example, you may have endpoints for registration, starting/finishing exams, retrieving data, or deleting registrations.

We'll put an example and you can check the source code to get full view over the project .

```go
func (s *Server) CreateRegister(c *fiber.Ctx) error {
	registration := &model.Registration{}
	accountId, _ := c.ParamsInt("account_id")
	examId, _ := c.ParamsInt("exam_id")

	if err := s.DB.Where("account_id = ? AND exam_id=?", accountId, examId).First(registration).Error; err ==
		return s.App.HttpResponseBadRequest(c, fmt.Errorf("User has already registerd for the exam"))
	}

	registration.IsCheated = false
	registration.Status = calculateRegStatus(registration)
	registration.AccountId = accountId
	registration.ExamId = examId
	tx := s.DB.Begin()
	err := tx.Create(registration).Error
	if err != nil {
		tx.Rollback()
		return s.App.HttpResponseInternalServerErrorRequest(c, err)
	}
	exam := &model.Exam{}

	if err = tx.Preload("Questions").First(exam, examId).Error; err != nil {
		tx.Rollback()
		return s.App.HttpResponseInternalServerErrorRequest(c, err)
	}
```

Figure 47-API-1

```go
func (s *Server) StartedExam(c *fiber.Ctx) error {

	registration := &model.Registration{}
	registrationId, _ := c.ParamsInt("id")

	if err := s.DB.Preload("Exam").First(registration, registrationId).Error; err != nil {
		return s.App.HttpResponseNotFound(c, fmt.Errorf("registiration not found"))
	}
	registration.Status = calculateRegStatus(registration)
	if registration.Status != EXAM_STATUS_NOT_STARTED {
		return s.App.HttpResponseForbidden(c, fmt.Errorf("the exam has started or your time is up"))
	}

	currentTime := time.Now()
	dd := registration.Exam.Duration * time.Minute
	currentTimeAfterDuration := currentTime.Add(dd)
	registration.StartTime = &currentTime
	registration.EndTime = &currentTimeAfterDuration
	registration.Status = calculateRegStatus(registration)

	err := s.DB.Save(registration).Error
	if err != nil {
		return s.App.HttpResponseInternalServerErrorRequest(c, err)
	}
	return s.App.HttpResponseOK(c, registration)
}
```

Figure 48-API-2

```go
func (s *Server) UserCheated(c *fiber.Ctx) error {
	registration := &model.Registration{}
	registrationId, _ := c.ParamsInt("id")

	if err := s.DB.First(registration, registrationId).Error; err != nil {
		return s.App.HttpResponseNotFound(c, fmt.Errorf("registiration not found"))
	}

	registration.Status = calculateRegStatus(registration)
	if registration.Status != EXAM_STATUS_STARTED {
		return s.App.HttpResponseForbidden(c, fmt.Errorf("the exam has not started yet or the exam is finished"))
	}

	registration.IsCheated = true
	registration.Status = calculateRegStatus(registration)
	s.DB.Save(registration)
	return s.App.HttpResponseOK(c, registration)
}
```

Figure 49-API-3

5.4 Tools and structure

Operating system :- Linux .

Backend language :- Golang.

Frontend framework :- React.

Database :- MySQL.

Cache Redis .

Docker to mange our containers.

5.5 Summary

In this chapter, we presented an introduction, followed by implementation of the database and summary about the structure.

6.0 System testing and installation

6.1 Introduction

In this chapter, we will test the system by a certified globally test called the Heuristic Evaluation Test and Cooperation Evaluation Test. This chapter presents the introduction in Section 6.1, heuristic evaluation is showed in Section 6.2 in addition to the cooperative evaluation test in Section 6.3, Section 6.4 presents the system installation. Furthermore, this chapter is finally summarized in Section 6.5.

6.2 Heuristic Evaluation

heuristic Evaluation seek to measure the usability of user interfaces in independent walkthroughs and report issues, this assessment is conducted by a small set of usability expert evaluators, who have examined the user interface systematically inspect a user interface and judge its compliance with a set of heuristics Such processes help prevent product failure. Heuristics are general rules that describe common properties of usable user interfaces. They help evaluators to focus their attention on specific aspects of user interfaces.

the Table Below shows the list of heuristics of usability evaluation and their descriptions.

Table 9-Heuristics of Usability Evaluation List and their Descriptions.

Numbering Scheme	Heuristics	Description
H1	**Visibility of system status**	Systems should always keep your users informed about what's going on screen "Where am I?" and "Where can I go next?" through appropriate feedback within reasonable time.
H2	**Match between system and the real world**	The system should speak the users' language so the information always should be presented in a natural way.
H3	**User control and freedom**	giving user the freedom to navigate and perform actions. The freedom to undo any accidental actions or redo.
H4	**Consistency and standards**	Users should not wonder for different words, situations, or actions mean the same task. All pages should be in the same theme.
H5	**Error prevention**	good error messages will help the user in preventing a problem from first occurring. Either check for them and present users with a

		confirmation option before they commit to the action.
H6	**Recognition rather than recall**	It is always recommended to suggest the user with a set of options than to let him remember and type the whole word. The goal is to minimize the application of user memory.
H7	**Flexibility and efficiency of use**	The Interface should be flexible transforming itself between a novice user and an advanced user.
H8	**Aesthetic and minimalist design**	Interfaces should be cleared of unnecessary elements and content that do not support the page goals and tasks.
H9	**Help users recognize, diagnose, and recover from errors**	Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.
H10	**Help and documentation**	If user looks for this means, something is not right with the interface. A great user interface lets the user navigate through its features without any documentation or training.

Table 10Summary of Violations by Severity Rating

Severity Rating	Frequency	Ratio (%)
0	37	82.2
1	5	11.1%
2	2	4.4%
3	0	0
4	1	2.2%
Total:	45	100%

6.3 Cooperative Evaluation

Cooperative Evaluation involves the user as an active participant in the assessment process. in which the user is encouraged to see himself as a collaborator in the evaluation rather than just a subject. It is utilized to help designers identify the most important improvements to consider with the minimum time and effort.

after we finished the design phase, we have test it with users in different ages and fields as our friends and our family members we gave them a brief about our application and then ask them to test some predefined use cases. Each participant was monitored, and each task was timed, here are the tables that show the cooperative evaluation summary for each participant.

Table 11-Participant1-as-auser

Task	Test	Time taken to complete the task
1	Sign up	15 seconds
2	Login	5 seconds
3	Review available exam	2 seconds
4	Select exam	1 second
5	Start exam	5 seconds
6	End exam	5 seconds
7	View result	1 hour
8	Download certification	20 seconds

Table 12-Participant2-as-auser

Task	Test	Time taken to complete the task
1	Sign up	30 seconds
2	Login	12 seconds
3	Review available exam	4 seconds
4	Select exam	2 second
5	Start exam	8 seconds
6	End exam	9 seconds
7	View result	1 hour
8	Download certification	20 seconds

6.3 Software Installation

The user need any browser to access our website .

6.4 Summary

This chapter showed a testing and evaluation for Mo3tamd System. The heuristic evaluation was conducted with 3 expert users. The Heuristic and cooperative evaluation have also shown competitive and acceptable performance for the system indicating that the system is easy to use and has less usability problems.

7.0 Project conclusion and future work

7.1 Introduction

in conclusion, Mo3tamad is an innovative solution designed to prevent cheating and provide valid certifications for enrolled exams. With its robust security measures, including exam monitoring.

Through thorough testing and evaluation, Mo3tamad has proven to be a reliable and user-friendly platform. The system's installation is straightforward, enabling seamless deployment for both administrators and users.

Mo3tamad opens up opportunities for individuals to enhance their qualifications and gain recognition for their accomplishments. It contributes to fair assessments and fosters trust in the education and professional sectors.

7.2 Overall Weaknesses

- We can’t control participant computer until we don’t have desktop application yet.
- Video recording doesn’t support yet for tracking participant during exam.
- Need more powerful UI/UX.

7.3 Overall Strengths

Mo3tamad boasts the following significant strengths:

- Global Company Collaboration: Mo3tamad fosters partnerships and collaborations with global companies, enabling individuals to access a wide range of prestigious certification exams from renowned international organizations.

- Robust Cheat Prevention System: Mo3tamad integrates a robust cheat prevention system, employing advanced techniques and technologies to detect and deter cheating behaviors, ensuring the authenticity and credibility of the exams.

- Creative and Innovative Approach: Mo3tamad introduces a creative and innovative approach to certification, leveraging technology and modern methodologies to provide individuals with a reliable and efficient means of obtaining certificates from reputable global companies.

7.4 Future Work

- Implement desktop application using C++ or Go to record user screen, log any transaction made by the user, and track user opened application during the exam.
- Add payment method on the website.
- Enhance UI/UX to make website more friendly to users.
- Use computer vision algorithm to track user behavior from camera during exam .
- Add more question types in the future .

7.5 Summary

In this chapter we talked about Mo3tamad weaknesses and strength points , also we talked about what we'll do in the future.